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PETER KENNEDY, Section Editor

Student Effort and Performance over the Semester

Gregory A. Krohn and Catherine M. O'Connor

Abstract: The authors extend the standard education production function and student time allocation analysis to focus on the interactions between student effort and performance over the semester. The purged instrumental variable technique is used to obtain consistent estimators of the structural parameters of the model using data from intermediate macroeconomics. The results of their study suggest that (1) students respond to higher midterm scores by reducing the number of hours they subsequently allocate to studying for the course; (2) contrary to results based on semester totals, class attendance is not related to examination scores throughout the semester; (3) study time has a small, but statistically significant, negative effect on student performance; and (4) although men outperform women on examinations, the difference may diminish over the course of the semester.

Key words: education production function, student effort
JEL code: I2

In this study, we examined the relationship between student effort, measured by hours spent on various course activities, and student performance over the semester in intermediate macroeconomics. We developed a theoretical model and obtained consistent estimates of the structural parameters in equations for student hours and performance. By considering the intrasemester interactions between student effort and performance, we gained some theoretical and empirical insights

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into their relationship that have not been discussed previously in the literature on economic education.

The measured effect of student effort on performance in economics courses found in past studies has been mixed. In an informative survey of research on economic education, Siegfried and Fels (1979) concluded that student effort did not seem to be related to performance on standardized tests (see also Becker 1997) but that class attendance seemed to be important for performance on locally constructed tests. More recent studies involving a variety of courses support the conclusion that class attendance positively affects student performance (Schmidt 1983—introductory macroeconomics; Lumsden and Scott 1987—introductory economics; Park and Kerr 1990—money and banking; Romer 1993—intermediate macroeconomics; Douglas and Sulock 1995—financial economics; Didia and Hasnat 1998—introductory finance). The effect of study time on performance was mixed. Schmidt (1983) and Douglas and Sulock (1995) found study time to be positively related to student performance. Borg, Mason, and Shapiro (1989) found study time to be insignificant overall among introductory macroeconomics students but positive for above average students and negative for below average students. On the other hand, Lumsden and Scott (1987), Gleason and Walstad (1988), and Park and Kerr (1990) found study time to be unrelated to student performance, and Didia and Hasnat (1998) found study time to be negatively related to performance.

Our analysis extends the standard education production function and student time allocation model (e.g., McKenzie and Staaf 1974; Needham 1978; Becker 1982, 1987; Becker and Rosen 1992; Allgood 2001) to consider interactions between student effort and performance over the semester. In our model, a student chose the time spent before each examination needed to maximize the utility received from the course grade and from other uses of time. Optimal examination scores were a function of hours of effort and ability, and optimally chosen hours depended on ability as well as on realized performance. One important prediction of the model was that the higher the score on a midterm examination, all else equal, the less time students spent subsequently in the course. We found some empirical evidence to support this notion, which suggests that estimates of the effect of student effort on performance using semester totals might be biased downward.

The use of intrasemester data allowed us to obtain consistent estimates of the structural parameters in hours and performance equations by the purged instrumental variable technique (Chamberlain 1977). In contrast to results using semester totals, no relation between class attendance and examination scores was found. One reason for this discrepancy was a possible bias resulting from the failure to correct for the endogeneity of hours of effort. A small, but statistically significant, negative effect of study time on performance was also found.

We also explored male-female differences in performance in intermediate macroeconomics. Previous studies of the role of gender in economic education generally found that men outperformed women in introductory economics courses but that learning rates in college were similar for men and women (Siegfried 1979). Heath (1989) presented evidence that the gender differences in empirical studies may understate the true differences because of sample selection bias. Robb and Robb (1999) found that gender differences in performance and

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continued study in economics past the introductory level were not influenced by the gender of the instructor. Attempts to explain the gender differences in student performance by Anderson, Benjamin, and Fuss (1994) were unsuccessful, but Lumsden and Scott (1987) found that although men performed better than women on multiple-choice tests, women performed better than men did on essay examinations. We found that men outperformed women in intermediate macroeconomics on examinations consisting mostly of short-answer problems, but the intrasemester data indicated that the difference may diminish over the course of the semester.

THEORY

The theoretical model we present here is a simplification of the model we estimated, but the features and results are useful for specifying and interpreting the empirical model.

Consider a student who is undertaking a course of study in which the final grade is a weighted average of two test scores. Test scores are a known function of the student's initial ability (human capital), time spent on the course before each examination, and an unobserved random variable with a mean of zero. The random variable might represent unpredictable factors affecting student performance that are specific to the student or that relate to the teacher's grading practices. The student's satisfaction (utility) is a function of the course grade and time not spent on the course. Time not spent on the course in question includes time spent on (*a*) other courses; (*b*) at paid employment; and (*c*) on recreation, sleeping, and so forth. The student is assumed to choose the time spent before each examination to maximize the utility from the course grade and from other uses of his or her time.

The assumption that the only value a student receives from a course is the grade received is admittedly restrictive. The student may value learning itself, or what is learned may have value in later courses or in a career. Although the course grade is not the only thing that might give satisfaction to a student, it is a readily available measure of knowledge of the subject matter. It seems reasonable to assume that the student cares about the grade received and will expend effort to increase the grade, realizing that time spent on the course takes away from other valuable uses of time. See Becker and Rosen (1992) and Allgood (2001) for similar approaches to the student decisionmaking process.

Combining all other uses of time into a single variable is restrictive and ignores factors such as the difficulty of the student's other courses or the student's financial situation—factors that may affect time spent on other courses or at paid employment. Nonetheless, that the student values other uses of time creates a tradeoff with time spent on the course in question, which is sufficient for our purposes. As a practical matter, data on other uses of time were not available for the students in the sample, so a richer theoretical specification could not be exploited in estimation anyway.

In mathematical form, the model can be expressed as follows:

Denote G—the course grade,
L—time not spent on the course,
U—the utility function,
e_1—score on exam 1,

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e_2—score on exam 2,
w—the weight on exam 1 in determining the course grade,
f_1—the grade function for exam 1,
f_2—the grade function for exam 2,
A—initial ability (given),
h_1—time spent before the first exam,
h_2—time spent between exam 1 and exam 2,
v_1—a random variable affecting the score on exam 1,
v_2—a random variable affecting the score on exam 2,
T_1—time available before the first exam, and
T_2—time available between exam 1 and exam 2.

The student's problem is

$$\max_{h_1, h_2} U(G, L) \tag{1}$$

subject to

$$G = w\, e_1 + (1 - w)\, e_2, \qquad 0 < w < 1, \tag{2}$$

$$e_1 = f_1(A, h_1, v_1), \qquad 0 < e_1 \leq 1.00, \tag{3}$$

$$e_2 = f_2(A, h_1, h_2, v_2), \qquad 0 < e_2 \leq 1.00, \tag{4}$$

$$L = T_1 + T_2 - h_1 - h_2, \quad \text{and} \tag{5}$$

$$h_j \leq T_j \quad \text{for } j = 1, 2. \tag{6}$$

Note that hours spent on the course before the first exam (h_1) can influence the grade on the second exam (e_2). This reflects the assumption that a student's ability might increase over the semester as the result of effort earlier in the term.

Specifying that the student's utility was a function of total time not spent on the course over the semester (L) implies that the marginal disutility of time devoted to the course was the same before and after the midterm exam. A more general specification would be to split L into $L_1 = T_1 - h_1$ and $L_2 = T_2 - h_2$ and enter these into the utility function separately. This would not, however, alter qualitatively the theoretical results, so the simpler specification was adopted to facilitate comparative static analyses.

The partial derivatives of U, f_1, and f_2 were positive. The second-order partial derivatives of the utility function was negative ($U_{GG} < 0$, $U_{LL} < 0$), and the cross partial derivative was assumed to be nonnegative ($U_{GL} \geq 0$) as would be the case with a Cobb-Douglas utility function or the more general constant-elasticity-of-substitution utility function.

The second-order partial derivatives of the grade functions, however, were not necessarily negative everywhere. Suppose that the graph of the grade received and time spent before the exam had the shape of a learning curve (Figure 1). For low amounts of effort, the slope of the grade function was increasing, and the second derivative was positive. But the slope of the grade function (the marginal effect on grades of time spent on the course) eventually decreased, and the second derivative would turn negative.

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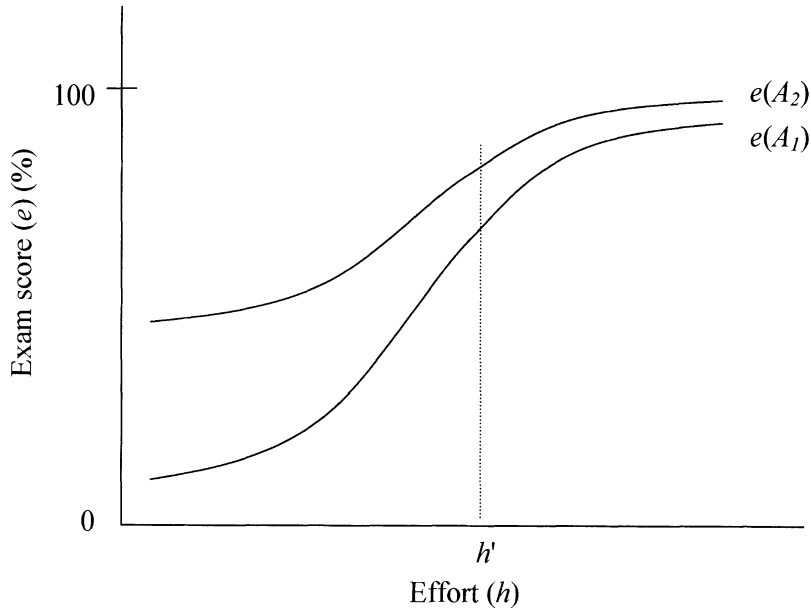

FIGURE 1. Exam score, effort, and ability.

The upper limit on exam grades had a noteworthy implication. An increase in initial ability would increase the grade received for any level of effort, but it could decrease the marginal effect of effort on the grade. An increase in initial ability from A_1 to A_2 would raise the grade schedule but at higher levels of time spent, such as h', the slope would get smaller (Figure 1). This implied that we might observe students with higher levels of initial ability having lower learning rates, if the learning rate was measured by marginal grade improvement.

One difficulty in solving the student's problem specified in equations (1) to (6) is that the variables v_1 and v_2 in the grade functions were unobserved before their respective examinations. One approach that we considered taking was to assume students maximize the expected utility of the course grade and alternative uses of time. If the utility function was quadratic and the constraints were linear, a closed-form solution would result, which would allow for comparative static analysis. However, two of the constraints, the grade functions, were inherently nonlinear, so the linear-quadratic model was not suitable.

In light of the nonlinearity in the grade functions, we adopted the following rather inelegant approach. The student's problem was split into two stages. First, the student chose hours, h_1, and provisionally chose hours, h_2, assuming that $v_1 = v_2 = 0$. Second, after the first exam, the student revised h_2 given v_1 and the score received on the first exam, assuming $v_2 = E(v_2 \mid v_1)$.

At the optimal amounts of effort, the student's marginal utilities from course time will equal the marginal utility of time spent on other activities. Sufficient

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conditions for the second order conditions to be satisfied are that the second-order partial derivatives of the grade functions are negative and that hours spent before the first exam have a relatively small effect on the learning rate of hours spent between the first and second exams. These conditions guarantee an interior solution in the choice of hours to spend on the course.

The optimal time spent on the course before the first exam can be written as an implicit function of the student's initial ability and parameters of the model such as w, the weight on the first exam.

$$h_1^* = h_1^*(A, w). \tag{7}$$

The optimal time spent on the course between the first and the second exams can be written as an implicit function of initial ability, model parameters, the score on the first exam, and the realization of the random variable, v_1.

$$h_2^* = h_2^*(A, w, e_1^*, v_1). \tag{8}$$

The exam scores realized depended on initial ability, the optimal time spent before each exam, and the random variables, v_1 and v_2.

$$e_1^* = \mathrm{f}_1(A, h_1^*, v_1), \quad \text{and} \tag{9}$$

$$e_2^* = \mathrm{f}_2\,(A, h_1^*, h_2^*, v_2). \tag{10}$$

The effects on effort and exam scores of initial ability, the unanticipated component of the score on the first exam, and the weight on the first exam can be analyzed with the relationships in equations (7) through (10).

Consider initial ability. By assumption, an increase in initial ability would increase both exam scores for given amounts of effort. But how does effort vary with initial ability? The answer depends on how initial ability affects learning rates (the marginal effect of time devoted to the course on grades). If an increase in initial ability does not increase the learning rate, then time spent on the course would decline. The reason is that an increase in ability would increase exam scores and the course grade, all else equal, and reduce the marginal utility of time spent on the course relative to the marginal utility of leisure.[1] On the other hand, if the marginal effect of effort on grades rises with initial ability, then spending more time on the course may increase the grade sufficiently to compensate for the loss of time spent on other activities. With the upper limit on course grades, this is more likely at low levels of initial ability.

An increase in the grade on the first exam resulting from an increase in the unobserved factor, v_1, would reduce effort on the second exam as intuition suggests if v_1 and v_2 are independent or positively correlated. An unanticipated increase in the grade on the first exam reduces the marginal utility of the course grade, which subsequently causes less time to be spent on the course and more time to be spent in other ways.

To our knowledge, one issue that has not been addressed in the literature on student effort and performance in economics is the effect on effort of changing the weights on the exams. This issue can be addressed in the present theory by changing the parameter w, the weight on the first exam. The effect on student effort of an

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increase in the weight on the first exam depends on the relative scores on the two exams. If the optimally determined scores on the two exams are equal given some value of w, then an increase in w would increase effort on the first exam and decrease effort on the second exam, as one might expect in a rational choice model. Furthermore, if the optimal score on the first exam is much less than the optimally determined score on the second, an increase in w could raise effort on the second exam as well as the first. The explanation is that, in this case, an increase in w would lower the course grade, which raises the marginal utility of time spent studying, even for the second exam. If, however, the optimally determined score on the first exam is larger than that on the second exam, an increase in w could lower the effort expended for both exams because the course grade would rise and its marginal utility would fall relative to the marginal utility of leisure. These results assume that the grade functions are known by the student and the instructor, which is a strong assumption. The lesson here is that teachers wanting to increase student effort on a course might consider increasing the weight on the more difficult exam and should beware of increasing the weight on the easier exam.

DATA AND PRELIMINARY EMPIRICAL RESULTS

We estimated an extension of the theoretical model presented earlier using data collected from students in three intermediate macroeconomics courses held at a private university in Pennsylvania: two in the fall semester of 1999 and one in the spring of 2000. The instructor and textbook were the same for each course, but the examinations were different in the spring semester. The textbook was first adopted in the fall of 1999, and as a result of that initial experience, some of the exercises and assignments were changed in spring 2000. A student's final grade was determined by scores on three exams (the first worth 10 percent; the second and third worth 20 percent each), a comprehensive final (30 percent), and homework assignments (20 percent). Students provided data on their grade point averages (GPA), the number of economics courses taken, and the grades received. The students' Scholastic Aptitude Test (SAT) scores, class year, and major were obtained from university records. The instructor classified students by gender.

Students also reported the number of hours spent each week preparing for and attending class. Students voluntarily kept records of their hours attending class, meeting with the professor, completing homework assignments, reading the textbook, reviewing their class notes, meeting with classmates, and seeing the teaching assistants (TAs). These records, or time sheets, were collected at several times during each semester to encourage record keeping. In the end, 85 out of 105 students turned in completed time sheets: 58 out of 70 in the fall and 27 out of 35 in the spring. Only one student dropped the course after the first week.

The number of students who did not provide complete data raises the possibility of selection bias when estimating the model of student effort and performance (see Montmarquette and Houle 1986; Douglas and Sulock 1995; Becker and Powers 2001). This does not appear to be a problem in the present study. Although 20 of the 105 students enrolled did not turn in completed time sheets, the missing data on student hours seemed to be random. Tests of the equality of

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TABLE 1. Sample Statistics

Demographic variable	All	Women	Men
Number of students	85	37	48
Juniors (%)	25	27	23
Seniors (%)	8	11	6
	Mean (St. Dev.)	*Mean (St. Dev.)*	*Mean (St. Dev.)*
Student characteristics			
SAT score	1,242 (103)	1,220 (103)	1,258 (101)
GPA	3.2 (0.4)	3.3 (0.3)	3.1 (0.5)
Economics GPA	3.3 (0.5)	3.4 (0.5)	3.2 (0.6)
Number of economics classes	1.6 (0.8)	1.8 (0.9)	1.5 (0.7)
Weekly student hours			
Attendance plus meeting professor	2.7 (0.3)	2.8 (0.2)	2.7 (0.3)
Completing assignments	1.1 (0.6)	1.1 (0.5)	1.1 (0.7)
Reading textbook	2.5 (1.4)	3.0 (1.7)	2.1 (1.0)
Reviewing notes	1.1 (0.7)	1.2 (0.8)	1.0 (0.5)
Meeting classmates	0.6 (0.7)	0.6 (0.7)	0.6 (0.7)
Meeting teacher assistants	0.1 (0.1)	0.1 (0.1)	0.1 (0.1)
Total	8.1 (2.6)	8.7 (2.8)	7.6 (2.3)
Student scores (%)			
Exam 1 (10)	75 (12)	74 (11)	75 (12)
Exam 2 (20)	80 (11)	79 (11)	80 (10)
Exam 3 (20)	81 (10)	82 (10)	80 (9)
Final exam (30)	80 (10)	79 (12)	81 (9)
Homework (20)	87 (9)	90 (7)	85 (9)
Course average	81 (7)	81 (8)	81 (6)

means of SAT scores, GPA, the number of economics classes, gender, and examination scores suggest that the students excluded from the sample did not differ significantly from the students included. In addition, an estimated probit model of the probability of turning in a complete time sheet revealed no significant effects of these explanatory variables, either individually or jointly.[2]

Sample statistics for variables in the dataset are provided in Table 1. The mean score on the SAT was 1,242. The mean GPA was 3.2, slightly lower than the average economics GPA of 3.3, although for many students this course was their second college economics course. A majority of students were sophomores and had not declared a major.

Students spent an average of 8.1 hours per week on this course: 2.7 hours per week in class and meeting with the professor and 5.4 hours per week studying (completing homework, reading the textbook, reviewing notes, and meeting with

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classmates and the TA).[3] The data also revealed that some students "crammed"; certain students reported zero hours for some weeks, whereas a large number of hours were reported for weeks just prior to an exam.

The mean course average was 81. Although not shown in the table, 11 students scored 90 or above, 37 scored in the 80s, 32 in the 70s, and five in the 60s.

Women comprised 44 percent of the students in the course. In comparison with men, women had lower SAT scores on average but slightly higher GPAs. Women tended to have taken more economics courses than the men had, perhaps because of the higher proportion of women who were juniors and seniors. One notable observation was that female students, on average, spent approximately one hour per week more on the course than did male students, primarily reading the textbook but also studying class notes. Women had the same mean course average score as men but tended to score lower on examinations and higher on homework.

Ordinary Least Squares Estimates Using Semester Totals Data

A preliminary analysis of semester totals was conducted to facilitate a comparison with previous results as well as to ascertain if our sample was atypical. We report ordinary least squares (OLS) estimates of equations using these data in Table 2. The equations for attendance (ha) and study (hs) hours had little, if any, explanatory power, although students enrolled in the fall semester appeared to have spent an average of 36 more hours studying during the entire semester, given their initial abilities.

The estimation results for the course average (CRSE) equations tended to support previous findings in the literature. Class attendance, for example, was positively related to performance in the course. Specifically, attending one more class was associated with an increase in a student's course average of 0.29 percentage point (column 3). Thus, a student who attended only a quarter of the 42 class meetings in a semester earned on average 9.1 percentage points less and one lower letter grade than a student with perfect attendance. Romer (1993) investigated the effects of attendance on performance in intermediate macroeconomics and found results of similar magnitude.

Study hours were found to have a small negative effect on student performance: one additional hour of studying reduced the course grade 0.04 percentage points per hour (column 3). When study hours were disaggregated (column 4), the only significant negative effect on performance came from studying class notes. One interpretation of this result is that students who did not understand the concepts in class might feel the need to study more but still performed poorly (Devadoss and Foltz 1996). If this were true, one would expect a negative coefficient on the time spent reading the textbook as well as on the time spent studying class notes but that was not what we found. Another possibility is that students' notes were flawed either through mistakes or omissions and the more they studied them, the more likely they were to gain misconceptions.

Positive effects on student performance were found for the human capital variables GPA and number of college economics courses taken (NECON). In a recent study of proxies for student aptitude, Grove and Wasserman (2003) found that some measure of grade point average is the best single proxy, but GPA combined

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TABLE 2. Ordinary Least Squares Coefficients from Models Using Semester Totals Data

Explanatory variable	Dependent variable			
	ha	*hs*	CRSE (a)	CRSE (b)
GPA	1.24	14.6	11.5**	11.7**
	(1.31)	(10.1)	(1.42)	(1.40)
SAT	−0.28	−4.66	0.79	0.001
	(0.40)	(3.83)	(0.53)	(0.006)
NECON	0.37	7.69	1.42*	1.81*
	(0.73)	(5.61)	(0.78)	(0.80)
FEMALE	1.28	10.7	−2.62*	−2.80*
	(1.10)	(8.47)	(1.18)	(1.22)
FALL99	1.83	36.0**	−1.75	−2.64
	(1.21)	(9.32)	(1.41)	(1.44)
ha			0.29*	0.35**
			(0.12)	(0.12)
hs			−0.04*	
			(0.02)	
hs—homework				−0.08
				(0.06)
hs—reading textbook				0.03
				(0.03)
hs—reviewing notes				−0.19**
				(0.06)
hs—classmates				−0.003
				(0.06)
hs—TA				−0.39
				(0.31)
R^2	0.07	0.22	0.54	0.59
p value	(0.318)	(0.001)	(0.000)	(0.000)

Note. CRSE = course average, as a percentage; *ha* = attendance hours; *hs* = study hours; NECON = number of econ courses taken; FEMALE = 1 if a female, 0 if a male; FALL99 = 1 if the course was taken in the fall semester 1999, 0 if taken in the spring semester 2000. Standard errors are in parentheses. Number of observations = 85. *P* values are given for *F* tests of the overall significance of each model.
*Statistically significant at the .05 Type I error level. **Statistically significant at the .01 Type I error level.

with SAT scores was even better. Here, however, the effect of SAT scores on student performance was insignificant.

Various studies have found that male economics students often outperform their female counterparts, especially on multiple-choice exams (e.g., Lumsden and Scott 1987; Anderson, Benjamin, and Fuss 1994). Statistically significant gender effects were found in the present study. The estimates in Table 2 suggest that the average woman's course average would be 2.6 to 2.8 percentage points lower than the average man's score, after controlling for initial ability and hours spent on the course.

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The results in Table 2 are suggestive only and might suffer from two problems. First, the examination scores and course averages were limited to the range of 0 to 100. OLS estimation of a model with a limited dependent variable can result in biased and inefficient estimators and invalid hypothesis tests (Greene 1997). We addressed this problem in our model by using intrasemester data with a transformation of student scores into an unlimited variable.

Second, there is the possibility of simultaneous equations bias. An endogenous variable in the system of equations, student hours, was an explanatory variable in the student performance equations. If there are unobservable factors that affect both student hours and performance, OLS coefficient estimators for the educational production function would be biased (Allison 1979; Schmidt 1983). We addressed the endogeneity problem in our model of intrasemester variables by estimating a system of equations with the purged instrumental variable technique.

OLS AND PURGED INSTRUMENTAL VARIABLE ESTIMATES USING INTRASEMESTER DATA

Consider a simple linear version of the optimal hours and exam scores functions given by equations (7) through (10) from the student maximization problem presented earlier. The system of structural equations for student i arising from this problem may be written

$$\begin{aligned} h_{1i} &= \boldsymbol{\beta}_1\mathbf{A}_i + v_{1i} \\ e_{1i} &= \gamma_{21}h_{1i} + \boldsymbol{\beta}_2\mathbf{A}_i + v_{2i} \\ h_{2i} &= \gamma_{31}h_{1i} + \gamma_{32}e_{1i} + \boldsymbol{\beta}_3\mathbf{A}_i + v_{3i} \\ e_{2i} &= \gamma_{41}h_{1i} + \gamma_{43}h_{2i} + \boldsymbol{\beta}_4\mathbf{A}_i + v_{4i}, \end{aligned} \tag{11}$$

where the endogenous variables hours of effort and exam scores for the two exams are denoted by the variables h_1, h_2 and e_1, e_2, respectively. The variable $\mathbf{A}_i$ denotes a vector of exogenous characteristics related to student i's ability and v_{gi}, in each of $g = 1, \ldots, 4$ equations, represents an unobserved disturbance. The vectors $\boldsymbol{\beta}_g$ and the γ's are parameters to be estimated.

According to the model, exam scores and hours decisions are determined by past or current outcomes, not future outcomes; the system given by equation (11) is recursive. Thus, despite the endogeneity of effort and exam scores, equation (11) could be consistently estimated by applying OLS to each equation separately as long as the v_g's are independent (Greene 1997).

The v_g's, however, may be correlated across equations because of unobserved factors not controlled for by included variables that affect effort and performance over the semester. For example, the v_g's would be correlated if student ability or tastes for macroeconomics are not perfectly controlled for by the exogenous variables included in $\mathbf{A}_i$. Additional unobserved factors found in the v_g's related to such things as the professor's teaching style, the kinds of course materials used, or the difficulty of assignments and exams, are also likely to affect effort and performance over the semester and cause correlation among the v_g's.[4]

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A more general specification for the disturbance term in equation (11), therefore, is an error components structure, in which the unobserved factors contained in v_g are decomposed into the following components: (1) factors that are specific to the student (such as tastes for macroeconomics), ω; (2) factors that are specific to a particular semester but common to all students (such as exam difficulty), α; and (3) factors that are purely random, u. Indexing semester by t, the disturbance terms may, therefore, be written

$$v_{git} = d_g f_{it} + u_{git} \qquad \text{for } g = 1,\ldots,4, \tag{12}$$

where

$$f_{it} = \alpha_t + \omega_{it}.$$

The semester-specific component α_t was assumed constant over students (i) but was independently and identically distributed across semesters (t), with mean zero and variance σ_α^2. The student-specific component ω_{it} varied over both i and t, with mean zero and variance σ_ω^2 and was assumed to be uncorrelated with α_t. The u_g's were independent across equations, semesters, and students and independent of α_t and ω_{it}.

With this specification, the v_g's were independent in the special case where the d_g's were equal to zero ($v_g = u_g$, for $g = 1,\ldots,4$), and estimation can be accomplished by OLS applied to each equation separately. If the d_g's were not equal to zero, the v_g's would be correlated across equations because the unobservable factors α_t and ω_{it} were common to all equations, and a different estimation procedure would be needed to obtain consistent estimators of the system's parameters. A standard two-stage least squares technique could be used if the system had the necessary number of exogenous variables that could be used as valid instruments. However, if all the exogenous variables in the data set affect every endogenous variable in the model, as is the case here with all elements of $\mathbf{A}_i$ included in each equation, an alternative method must be used to identify and consistently estimate the system of equations with correlated errors.

One alternative that can be used is a purged instrumental variable (IV) technique (Chamberlain 1977) that exploits the error-components formulation of the disturbance terms. With the purged IV technique, instruments for the endogenous explanatory variables are formed by purging these variables of the influence of the common factor $f = \alpha_t + \omega_{it}$, using a pseudo two-stage technique. Briefly, the purged IV technique proceeds as follows. In the first stage, the first endogenous variable of the system, h_1 for the case given by equation (11), is used as a proxy for the factor f in the reduced form equations for the other endogenous variables of the system.[5] Residuals from the estimated reduced form equations are formed. The residuals are the endogenous variables purged of the influence of the factor f. In the second stage, these residuals are used as instruments for the endogenous variables in the structural equations of the system. Each instrument will be correlated with the endogenous variable it replaces and uncorrelated with the disturbance term of the relevant equation. Although consistent estimates are obtained, the purged IV method is likely to be inefficient because of possible weak correlation between the instruments and the endogenous variables. For complete details, see Hsiao (1986).[6]

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The model estimated was an expansion of the system given by equation (11). The semester was divided into time periods by four, not two, examinations, and the hours and homework scores (p) were measured accordingly.[7] Following the literature, hours data were disaggregated to isolate the effects of attendance hours (ha) from study hours (hs). Homework scores were modeled in a similar way as examination scores. We explicitly included homework scores to best characterize the decisions facing students in intermediate macroeconomics; homework scores were worth 20 percent of a student's final grade and might influence time allocation decisions. Because homework assignments were designed to help prepare students for examinations, we also allowed contemporaneous homework scores to determine examination scores in the estimated model.[8] The human capital, or initial ability, variables included in the model were GPA, SAT scores, and the number of college economics courses taken (NECON).[9] We also included dummy variable for student gender (FEMALE).

To address the limited range of the examination and homework scores and the associated nonlinear functional relationships, we transformed scores as follows:

$$e_j' = \ln [e_j/(1 - e_j)], \qquad j = 1, 2, 3, \text{ and } 4, \text{ and} \tag{13}$$

$$p_j' = \ln [p_j/(1 - p_j)], \tag{14}$$

where e_j is the decimal score on examination j, and p_j is the average of decimal homework scores before examination j but after examination $j - 1$. A few students earned perfect scores on some of the homework, and three students earned perfect scores on the third exam. These were recoded as 0.999 prior to transformation; there were no zeros scored.[10] The linear equations for the transformed variables in the recursive model imply that e_j and p_j are logistic functions of the explanatory variables, and the resulting function relating exam scores and student effort takes the shape of the learning curve shown in Figure 1.

Results from estimating the expanded system, equation by equation using OLS, are reported in Tables 3 and 4. We found using a Breusch-Pagan test of independence that we could reject the null hypothesis that the error terms were uncorrelated across equations, which suggests that the OLS estimators were biased. We reestimated the model using the purged IV method and report the results in Tables 5 and 6. Although not reported, some of the purged IV estimates of the d_g's, the coefficients on the common factor f in each equation, were statistically significant, providing further support for the conclusion that the OLS estimators were biased. In the following discussion of the purged IV results, however, we note some of the differences and similarities with the OLS results.

The purged IV estimates of the intrasemester equations indicated that hours in attendance and hours spent studying were largely unexplained with one notable exception. There is evidence in accordance with the present theory that students respond to higher exam scores by subsequently studying less (Table 5). To interpret the estimated coefficients, the marginal effects of an increase in exam scores on study time are reported at the bottom of Table 5. For each additional percentage point on the first exam, the average student spent roughly 0.5 fewer hours studying before the second exam, 0.3 fewer hours before the third exam, and 0.2

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TABLE 3. Ordinary Least Squares Coefficients for Hours Equations

Explanatory variable	Dependent variable							
	ha_1	ha_2	ha_3	ha_4	hs_1	hs_2	hs_3	hs_4
GPA	0.48	−0.75	−0.57	1.06*	4.15	9.68*	8.92*	5.61*
	(0.52)	(0.69)	(0.62)	(0.53)	(3.68)	(4.33)	(3.45)	(2.92)
SAT	0.13	−0.35	−0.04	−0.11	−1.31	−0.94	0.38	−0.36
	(0.20)	(0.25)	(0.21)	(0.17)	(1.41)	(1.58)	(1.17)	(0.95)
NECON	0.05	−0.45	−0.11	0.39	−0.04	0.85	3.57*	1.64
	(0.26)	(0.35)	(0.29)	(0.24)	(1.86)	(2.18)	(1.60)	(1.32)
FEMALE	0.18	1.66**	−0.10	0.11	4.36	1.74	−0.49	1.06
	(0.43)	(0.54)	(0.45)	(0.37)	(3.07)	(3.43)	(2.55)	(2.06)
e_1'		0.20	−0.09	−0.21		−7.89**	−7.42**	−5.18*
		(0.38)	(0.34)	(0.29)		(2.42)	(1.89)	(1.59)
e_2'			0.67	0.48			−2.53	0.49
			(0.39)	(0.32)			(2.17)	(1.77)
e_3'				−0.17				−0.46
				(0.19)				(1.04)
p_1'		−0.43	0.45	−0.11		−0.52	1.44	0.63
		(0.26)	(0.23)	(0.19)		(1.64)	(1.26)	(1.03)
p_2'			0.09	−0.06			−1.18	−0.60
			(0.16)	(0.13)			(0.87)	(0.72)
p_3'				0.09				0.55
				(0.10)				(0.52)
R^2	0.02	0.14	0.10	0.16	0.06	0.18	0.28	0.23
p value	(0.78)	(0.07)	(0.39)	(0.20)	(0.25)	(0.02)	(0.00)	(0.03)

Note. SAT scores are in 100s. $e_j' = \ln[e_j/(1 - e_j)]$, where e_j is the decimal score on examination j (e.g., 0.84). Similarly for the homework assignments, $p_j' = \ln[p_j/(1 - p_j)]$. The marginal effects of examination scores were evaluated at the sample means. Standard errors are in parentheses. Number of observations = 85. NECON = number of economics courses taken.
*Statistically significant at the .05 Type I error level. **Statistically significant at the .01 Type I error level.

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TABLE 4. Ordinary Least Squares Coefficients for Examination Equations

Explanatory variable	Dependent variable			
	e_1'	e_2'	e_3'	e_4'
GPA	0.83**	0.92**	1.28**	0.97**
	(0.18)	(0.15)	(0.28)	(0.19)
SAT	0.09	0.09	0.18*	0.15*
	(0.07)	(0.06)	(0.10)	(0.07)
NECON	0.35**	0.07	0.44**	0.06
	(0.09)	(0.08)	(0.14)	(0.09)
FEMALE	−0.31**	−0.30**	−0.02	−0.37*
	(0.15)	(0.13)	(0.23)	(0.15)
ha_1	0.006	0.048	0.015	−0.005
	(0.04)	(0.036)	(0.064)	(0.042)
ha_2		0.037	0.053	0.078*
		(0.028)	(0.052)	(0.034)
ha_3			−0.055	0.038
			(0.062)	(0.040)
ha_4				0.000
				(0.049)
hs_1	−0.015**	−0.013*	−0.006	0.011
	(0.006)	(0.006)	(0.010)	(0.007)
hs_2		−0.009*	0.053	0.004
		(0.005)	(0.052)	(0.007)
hs_3			−0.035**	−0.012
			(0.013)	(0.008)
hs_4				0.009
				(0.010)
p_1'	0.016			
	(0.075)			
p_2'		0.012		
		(0.043)		
p_3'			−0.031	
			(0.059)	
p_4'				0.026
				(0.035)
Constant	−2.72	−3.12	−4.90	−4.35
	(1.15)	(1.00)	(1.83)	(1.19)
R^2	0.34	0.44	0.41	0.41
p value	(0.00)	(0.00)	(0.00)	(0.00)

Note. SAT scores are in 100s. $e_j' = \ln[e_i/(1 - e_i)]$, where e_j is the decimal score on examination j (e.g., 0.84). Similarly, $p_j' = \ln[p_j/(1 - p_j)]$. Standard errors are in parentheses. Number of observations = 85. NECON = number of economics courses taken.
*Statistically significant at the .05 Type I error level. **Statistically significant at the .01 Type I error level.

fewer hours before the final exam. Over the course of the semester, the effects of exam scores on study time got smaller and less significant. The purged IV results did not show a significant GPA effect on study hours as was found using OLS (Table 3).

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TABLE 5. Purged IV Coefficients for Hours Equations

Explanatory variable	Dependent variable							
	ha_1	ha_2	ha_3	ha_4	hs_1	hs_2	hs_3	hs_4
GPA	0.46	−0.65	0.06	1.14**	4.10	3.48	0.62	1.70
	(0.52)	(0.62)	(0.52)	(0.42)	(3.70)	(3.87)	(3.10)	(2.45)
SAT	0.14	−0.21	−0.09	−0.11	−1.26	−1.63	−0.87	−1.21
	(0.20)	(0.24)	(0.20)	(0.16)	(1.41)	(1.48)	(1.18)	(0.93)
NECON	0.06	−0.47	0.06	0.31	0.02	−2.07	0.50	0.24
	(0.26)	(0.32)	(0.26)	(0.22)	(1.87)	(1.96)	(1.57)	(1.25)
FEMALE	0.17	1.46**	−0.18	0.06	4.30	4.52	3.70	3.41
	(0.43)	(0.52)	(0.43)	(0.35)	(3.09)	(3.23)	(2.58)	(2.04)
e_1'		0.18	0.08	−0.17		−8.58**	−5.51**	−3.60*
		(0.39)	(0.34)	(0.28)		(2.42)	(2.04)	(1.63)
e_2'			0.50	0.42			−2.28	−0.16
			(0.38)	(0.31)			(2.27)	(1.81)
e_3'				−0.21				0.06
				(0.19)				(1.12)
p_1'		−0.39	0.36	−0.11		0.46	1.16	0.49
		(0.28)	(0.23)	(0.19)		(1.71)	(1.36)	(1.09)
p_2'			0.17	−0.03			−0.49	0.04
			(0.34)	(0.13)			(0.93)	(0.76)
p_3'				0.05				0.27
				(0.10)				(0.59)

(table continues)

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TABLE 5. Continued

Explanatory variable	Dependent variable							
	ha_1	ha_2	ha_3	ha_4	hs_1	hs_2	hs_3	hs_4
R^2	0.02	0.13	0.08	0.15	0.07	0.19	0.15	0.13
p value	(0.85)	(0.09)	(0.55)	(0.23)	(0.34)	(0.01)	(0.11)	(0.34)
Addenda: Marginal effects of examination scores								
e_1						−45.5	−29.3	−19.1
e_2							−12.1	−0.9
e_3								0.3

Note. SAT scores are in 100s. $e_j' = \ln[e_j/(1 - e_j)]$, where e_j is the decimal score on examination j (e.g., 0.84). Similarly for the homework assignments, $p_j' = \ln[p_j/(1 - p_j)]$. The marginal effects of examination scores were evaluated at the sample means. Standard errors are in parentheses. Number of observations = 85. NECON = number of economics courses taken.
*Statistically significant at the .05 Type I error level. **Statistically significant at the .01 Type I error level.

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TABLE 6. Purged IV Coefficients for Examination Equations

Explanatory variable	Dependent variable			
	e_1'	e_2'	e_3'	e_4'
GPA	0.78**	0.92**	1.30**	0.99**
	(0.18)	(0.16)	(0.28)	(0.19)
SAT	0.11	0.13*	0.21*	0.15*
	(0.07)	(0.06)	(0.10)	(0.08)
NECON	0.36**	0.12	0.38**	0.01
	(0.09)	(0.08)	(0.14)	(0.09)
FEMALE	−0.37**	−0.37**	−0.05	−0.34*
	(0.15)	(0.13)	(0.23)	(0.15)
ha_1	.004	−0.029	−0.087	−0.004
	(0.039)	(0.034)	(0.059)	(0.039)
ha_2		0.022	0.028	0.058
		(0.030)	(0.054)	(0.036)
ha_3			−0.050	0.012
			(0.064)	(0.043)
ha_4				0.026
				(0.054)
hs_1	−0.015**	0.007	−0.005	0.000
	(0.006)	(0.005)	(0.008)	(0.006)
hs_2		−0.012**	0.013	0.001
		(0.005)	(0.10)	(0.006)
hs_3			−0.033**	−0.010
			(0.013)	(0.009)
hs_4				0.004
				(0.010)
p_1'	0.044			
	(0.078)			
p_2'		0.020		
		(0.046)		
p_3'			−0.005	
			(0.063)	
p_4'				0.025
				(0.037)
Constant	−3.00	−2.82	−4.50	−3.28
	(1.05)	(1.11)	(1.91)	(1.28)
R^2	0.35	0.36	0.37	0.34
p value	(0.00)	(0.00)	(0.00)	(0.02)

Note. SAT scores are in 100s. $e_j' = \ln[e_i/(1 - e_i)]$, where e_j is the decimal score on examination j (e.g., 0.84). Similarly, $p_j' = \ln[p_j/(1 - p_j)]$. Standard errors are in parentheses. Number of observations = 85. NECON = number of economics courses taken.
*Statistically significant at the .05 Type I error level. **Statistically significant at the .01 Type I error level.

We show the purged IV estimates of the examination equations in Table 6.[11] The results indicate that the human capital variables GPA, SAT, and NECON are significantly and positively related to exam scores. Recall that SAT scores were not statistically significant in equations estimated using semester totals. The

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marginal effects of these variables evaluated at sample means are shown in Table 7. An additional point in the GPA was predicted to increase exam scores by 6 to 24 percentage points (0.06 to 0.24) or by somewhat more than one letter grade, on average. An additional 100 points on SAT scores was predicted to increase exam scores by 1 to 4 percentage points (0.01 to 0.04). Having taken an additional economics course increased scores on the first and third examinations by 7 percentage points (0.07).

Unlike the OLS results using semester totals data, the purged IV results did not indicate that attendance significantly affected examination scores (Table 6). This might be caused by less variability in hours of attendance between examinations than over the semester as a whole. Another factor might be a bias in the OLS estimates. OLS estimates of the four exam equations (Table 4) indicated coefficients on attendance hours that were larger for the most part than the IV estimates but still not statistically significant.

The purged IV estimates of the first three examination equations confirmed the negative relationship between study hours and exam scores observed with the semester totals data (Table 6), but looking at the marginal effects of study hours on exam scores in Table 7, we find them to be approximately 1/10th the size of the OLS estimates using semester totals (−0.003 compared with −0.04). Part of this difference might reflect a downward (upward in absolute value) bias in the semester total estimates because of the negative effect of early exam scores on subsequent study time found in the hours equations.

Similar to previous studies of students in other economics courses, we found that men outperformed women in intermediate macroeconomics (Table 6). Why men outperformed women in economics after controlling for ability and effort remains a puzzle. An interesting pattern, however, can be observed over the

TABLE 7. Marginal Effects on Examination Scores

Explanatory variable	e_1	e_2	e_3	e_4
GPA	0.148	0.118	0.241	0.064
SAT	0.020	0.017	0.040	0.010
NECON	0.067	0.015	0.070	0.001
ha_1	0.001	−0.004	−0.016	−0.000
ha_2		0.003	0.005	0.004
ha_3			−0.008	0.001
ha_4				0.002
hs_1	−0.003	0.001	−0.001	0.000
hs_2		−0.002	0.002	0.000
hs_3			−0.006	−0.001
hs_4				0.000

Note. Marginal effects are calculated for the purged IV results reported in Table 6 and evaluated at sample means. Examination scores are in decimals (e.g., 0.84). SAT scores are in 100s. Estimates are for men. Estimates for women are proportionately larger in absolute value. NECON = number of economics courses taken.

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TABLE 8. Predicted Values of Male and Female Examination Scores

Examination	Men	Women	Difference
e_1	0.745	0.667	0.078
e_2	0.848	0.795	0.053
e_3	0.754	0.743	0.011
e_4	0.930	0.904	0.026

Note. Calculated for the purged IV results reported in Table 6 and evaluated at full sample means.

semester in the present study. The difference between the predicted values of male and female examination scores declined from 8 percentage points (0.078) on the first exam to 5 percentage points on the second exam (0.053) to 1 percentage point on the third exam (0.011) (Table 8). The difference on the comprehensive final exam rose slightly to nearly 3 percentage points (0.026). Although previous studies have not found different learning rates between men and women (Siegfried 1979), our study suggests that the male-female gap can be narrowed over the semester, although not simply by women spending more time studying. It is possible, of course, that the declining gender effect we found may have resulted from the nature of the particular exams and not be a general phenomenon.

To check the robustness of our results, we reestimated the model on untransformed data using the purged IV technique. Because we wanted to continue to capture the nonlinearities inherent in the grade function, we interacted hours data with GPA and included these new variables in the examination and homework equations to account for any possible differences in learning rates across students of different abilities. Estimated coefficients for hours and examination equations are reported in Tables 9 and 10, respectively.

Results for the untransformed data tended to support those found using transformed data. Students were found to respond to higher exams scores by spending fewer hours studying, and this effect got smaller and less significant over the course of the semester (Table 9). Attendance hours were not found to have a significant effect on exam scores, except for exam 3, where the effect was positive for lower ability (GPAs < 3.00) students (Table 10). We also found the negative relationship between study hours and exam scores for the average student in the sample, but extra study hours appeared to be beneficial for students with lower GPAs, at least on the first two exams (Table 10). These results tend to offer some support for the learning curve relationship between effort and performance shown by Figure 1, which indicates that the decreasing marginal effects of effort may cause strong students to appear to have low learning rates. The untransformed data also suggest that the gender effect was smaller for the third exam compared to the first two exams and not statistically significant but was about the same for the final as for the first two tests (Table 10).

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TABLE 9. Purged IV Coefficients for Hours Equations Using Untransformed Data

Explanatory variable	Dependent variable							
	ha_1	ha_2	ha_3	ha_4	hs_1	hs_2	hs_3	hs_4
GPA	0.58	−0.66	0.02	1.18**	5.28	3.48	0.36	1.15
	(0.52)	(0.56)	(0.56)	(0.40)	(3.58)	(3.37)	(3.12)	(2.58)
SAT	0.18	−0.21	−0.05	−0.14	−0.79	−1.63	0.97	−0.21
	(0.20)	(0.27)	(0.18)	(0.14)	(1.37)	(1.68)	(1.08)	(0.99)
NECON	0.09	−0.47	−0.04	0.27	0.36	−2.07	2.84	1.02
	(0.26)	(0.30)	(0.28)	(0.21)	(1.80)	(2.55)	(2.14)	(1.85)
FEMALE	0.31	1.46*	−0.01	0.13	5.81*	4.52	−2.94	−0.28
	(0.43)	(0.56)	(0.45)	(0.37)	(3.01)	(3.46)	(3.01)	(2.67)
e_1		1.36	0.81	0.34		−58.5**	−17.2	−11.6
		(2.87)	(1.84)	(1.51)		(19.6)	(10.5)	(8.52)
e_2			1.92	0.50			−26.6*	−10.6
			(1.78)	(1.47)			(10.2)	(8.31)
e_3				0.89				7.24
				(2.52)				(14.2)
p_1		−1.25	2.33	−0.81		16.5	2.30	5.89
		(1.79)	(1.84)	(1.72)		(10.5)	(10.5)	(9.65)
p_2			2.43	1.36			0.24	6.84
			(2.03)	(1.78)			(11.6)	(10.0)
p_3				0.32				−1.08
				(1.03)				(5.82)
R^2	0.07	0.12	0.07	0.13	0.14	0.19	0.21	0.17
p value	(0.45)	(0.13)	(0.65)	(0.38)	(0.03)	(0.01)	(0.02)	(0.16)

Note. Standard errors are in parentheses. NECON = number of economics courses taken.
*Statistically significant at the .05 Type I error level. **Statistically significant at the .01 Type I error level.

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TABLE 10. Purged IV Coefficients for Examination Equations Using Untransformed Data

Explanatory variable	Dependent variable			
	e_1	e_2	e_3	e_4
GPA	0.121**	0.270**	−0.029	0.160*
	(0.03)	(0.07)	(0.19)	(0.06)
SAT	0.023*	0.022**	0.011	0.015
	(0.01)	(0.01)	(0.01)	(0.01)
NECON	0.055**	0.008	0.040**	0.001
	(0.01)	(0.01)	(0.01)	(0.01)
FEMALE	−0.050*	−0.050*	−0.004	−0.054*
	(0.02)	(0.02)	(0.02)	(0.02)
ha_1	0.062	−0.007	−0.001	0.001
	(0.06)	(0.01)	(0.00)	(0.01)
ha_2		0.053	0.004	0.009
		(0.05)	(0.00)	(0.01)
ha_3			0.068*	0.002
			(0.04)	(0.01)
ha_4				−0.045
				(0.06)
hs_1	0.016*	0.001	−0.000	0.000
	(0.01)	(0.00)	(0.00)	(0.00)
hs_2		0.013*	0.001	0.000
		(0.01)	(0.00)	(0.00)
hs_3			0.001	−0.001
			(0.01)	(0.00)
hs_4				0.008
				(0.01)
p_1	0.098			
	(0.09)			
p_2		−0.011		
		(0.09)		
p_3			0.004	
			(0.04)	
p_4				0.05
				(0.07)
GPA*ha_1	−0.018			
	(0.02)			
GPA*ha_2		−0.015		
		(0.01)		
GPA*ha_3			−0.023*	
			(0.01)	
GPA*ha_4				0.014
				(0.02)
GPA*hs_1	−0.006*			
	(0.00)			
GPA*hs_2		−0.004*		
		(0.00)		

(table continues)

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TABLE 10. Continued

Explanatory variable	Dependent variable			
	e_1	e_2	e_3	e_4
GPA*hs_3			−0.001	
			(0.00)	
GPA*hs_4				−0.002
				(0.00)
R^2	0.41	0.37	0.50	0.30
p value	(0.00)	(0.00)	(0.00)	(0.02)

Note. Standard errors are in parentheses. NECON = number of economics courses taken. *Statistically significant at the .05 Type I error level. **Statistically significant at the .01 Type I error level.

Our purged IV results did not differ dramatically from those obtained by estimating each equation separately by OLS, suggesting that the bias we corrected for was small. Because we analyzed data for two semesters from the same professor, this result may be true for any omitted semester-specific effects (α_t's). The advantages of using the purged IV method are likely to be more pronounced if we had data on additional semesters or data from different professors teaching the same course.

CONCLUSION

This study fills a gap in the economic education literature by extending the standard education production function and student time allocation analysis to focus on the interaction of student effort and performance over the course of a semester. The structural parameters of the resulting system of equations were consistently estimated by using the purged instrumental variable technique. The study analyzed data from intermediate macroeconomics, a required course for economics majors, on which few studies have focused.

Our findings using intrasemester data shed some light on the relationship between student effort and performance and suggest avenues for further research. Our results indicate that students respond to higher midterm exam scores by reducing the number of hours they subsequently allocate to studying for the course. This implies that estimates of the effect of study time on performance formed from semester totals data might be biased downward. Our interpretation of this finding is that a higher midterm score reduces the marginal utility of studying relative to the marginal utility of leisure. Another interpretation that lies outside the scope of the present theory is that students do not know the grade functions at the beginning of the semester, and they study excessively until they discover what they need to study to receive the grade they desire. An investigation into this interpretation would be welcome.

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Our results call into question the findings in other studies that attendance affects performance on locally constructed tests. Although attendance is positively related to student performance in the semester totals data, no relationship is found between attendance and scores on individual examinations using the intrasemester data.

We also found, as others have, a small, statistically significant, negative effect of study time on exam scores. This puzzling result suggests that the ability to learn in an economics course may not be adequately captured by commonly employed measures of human capital or, more fundamentally, that the education production function approach to learning is misspecified. A related finding is that SAT scores have a statistically significant effect on student performance in the intrasemester equations but not in the semester totals equations. The reason for the difference is not clear, but one implication is that studies using semester totals that have not found SAT scores to have significant effects when other measures of human capital are included ought to be interpreted cautiously.

Finally, the intrasemester data suggest that the male-female difference in performance in intermediate macroeconomics might diminish over the semester. Verification of this result is needed before it can be considered a firm finding. In any case, the result makes us wonder if the gender differences found in introductory and intermediate economics courses narrows or even disappears as more economics courses are taken. Studies of upper-level undergraduate courses, graduate entrance examinations, and graduate courses could address the question of the persistence of student gender differences in economics over a program of study.

NOTES

1. It is assumed that the marginal utility of grades is independent of initial ability. If the marginal utility of grades were to rise with initial ability, then an increase in effort could result.
2. Estimation was performed using STATA.
3. It was apparent that some students reported attendance at review sessions held before the exams as attending class, others as meeting with the professor. These two categories were combined into one for the analysis of the data. Thus, attendance hours comprise both class attendance and meetings with the professor.
4. The importance of course-specific effects is suggested by Allison (1976), Becker and Powers (2001), and Lumsden and Scott (1987).
5. The disturbances of the reduced form equations are functions of the u_{it}'s (in particular u_{1it}). As a result, the h_{1it}'s will be correlated with the reduced form disturbances. A legitimate instrument must, therefore, be found for the h_{1it}'s in this first stage of the purged IV method. As suggested by Chamberlain (1977), using different members of the same group provides a legitimate instrument. In our case, this meant using h_{1jt} , $j \neq i$ for h_{1it} (that is using different students in the same semester). The h_{1jt}'s will be uncorrelated with the reduced form disturbances and will be correlated with the h_{1it}'s provided that the variance of the semester-specific error component is not equal to zero and h_1 is a function of the factor f. Under the assumptions of the empirical model, any method of choosing the h_{1jt}'s will yield consistent estimates. We chose these by using adjacent students in an alphabetical listing of the students in each semester.
6. Identification of the system relies on the restrictions placed on the disturbance covariance matrix by the error structure of the system. In addition, a necessary condition for identification is that one of the endogenous explanatory variables in the recursive system be excluded. The previous theory discussion suggests that exam scores are a function of current and past hours decisions, not past exam scores, so this condition is met in our case.
7. Time periods between exams were fairly uniform over the semester, with the exception of the time before the final exam, which was shorter. In any case, the marginal effects of effort on exam scores need not be affected by differences in the lengths of the time periods between exams.

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8. Students were allowed to help one another on homework. In terms of the model, any unobserved "study group" effect for a particular student can be thought of as part of the ω_{it} component of the v_g's.
9. Recall that most students in the class were sophomores who had not declared their majors and for whom intermediate macroeconomics was their second college economics course. We included the number of economics courses taken as a proxy for a student's initial knowledge of economics.
10. Recoding perfect scores as 0.99 or 0.9999 prior to transformation did not substantially alter estimation results.
11. Equations for homework scores were estimated but not reported. Most of the coefficients were insignificant.

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